Midterm Task 3 Normalization using Power Query

1. The data - uncleaned

a.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
order_id	date	product_id	product_name	price	currency	category_id	category_name	subcategory_id	subcategory_name	store_id	store_name	city_id	city_name	country_id	country
GMT-36520	01/02/2020	P002	EconoTread	699.99	GBP	C01	Cardio Equipment	SC01	Treadmills	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GMT-36520	01/02/2020	P020	Door Anchor Straps	24.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GMT-36520	01/02/2020	P012	Olympic Flat Bench	179.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GMT-36520	01/02/2020	P012	Olympic Flat Bench	179.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GMT-36520	01/02/2020	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
MOF-12749	24/07/2020	P004	Recumbent Cycler	599.99	USD	C01	Cardio Equipment	SC02	Exercise Bikes	S002	Muscle Mart	LA	Los Angeles	US	United States
MOF-12749	24/07/2020	P003	SpinMaster Bike	799.99	USD	C01	Cardio Equipment	SC02	Exercise Bikes	S002	Muscle Mart	LA	Los Angeles	US	United States
MOF-12749	24/07/2020	P016	Heavy Duty Resistanc	39.99	USD	C03	Functional Fitness	SC08	Resistance Bands	S002	Muscle Mart	LA	Los Angeles	US	United States
VKL-97451	19/12/2022	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GJO-36640	14/04/2022	P019	Pro Suspension Trai	129.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GJO-36640	14/04/2022	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GJO-36640	14/04/2022	P017	Cast Iron Kettlebell 1	59.99	GBP	C03	Functional Fitness	SC09	Kettlebells	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GJO-36640	14/04/2022	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
GNN-19269	02/10/2022	P021	Recovery Roller Dail	39.99	USD	C04	Recovery and Wellne	SC11	Foam Rollers	S001	FitGear Hub	NY	New York	US	United States
GNN-19269	02/10/2022	P023	PulseRelief Massage	199.99	USD	C04	Recovery and Wellne	SC12	Massage Guns	S001	FitGear Hub	NY	New York	US	United States
GNN-19269	02/10/2022	P020	Door Anchor Straps	24.99	USD	C03	Functional Fitness	SC10	Suspension Trainers	S001	FitGear Hub	NY	New York	US	United States
GNN-19269	02/10/2022	P011	Adjustable Bench Pr	249.99	USD	C02	Strength Training	SC06	Weight Benches	S001	FitGear Hub	NY	New York	US	United States
TFA-40365	25/10/2021	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
TFA-40365	25/10/2021	P022	Trigger Point Grid Ro	59.99	GBP	C04	Recovery and Wellne	SC11	Foam Rollers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
TFA-40365	25/10/2021	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
TFA-40365	25/10/2021	P011	Adjustable Bench Pr	249.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
AQN-95165	16/01/2021	P015	FlexForce Band Kit	49.99	USD	C03	Functional Fitness	SC08	Resistance Bands	S003	Strength Central	CH	Chicago	US	United States
PSG-64239	16/06/2020	P001	Pro Runner 3000	1299.99	GBP	C01	Cardio Equipment	SC01	Treadmills	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
PSG-64239	16/06/2020	P020	Door Anchor Straps	24.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
PSG-64239	16/06/2020	P016	Heavy Duty Resistanc	39.99	GBP	C03	Functional Fitness	SC08	Resistance Bands	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
OAT-63948	25/07/2023	P018	Adjustable Kettlebell	149.99	USD	C03	Functional Fitness	SC09	Kettlebells	S003	Strength Central	CH	Chicago	US	United States
OAT-63948	25/07/2023	P025	Premium Yoga Mat	69.99	USD	C04	Recovery and Wellne	SC13	Yoga Mats	S003	Strength Central	CH	Chicago	US	United States
IAY-77502	05/02/2020	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S004	Elite Gym Supply	LO	London	UK	United Kingdom
IAY-77502	05/02/2020	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S004	Elite Gym Supply	LO	London	UK	United Kingdom
IAY-77502	05/02/2020	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S004	Elite Gym Supply	LO	London	UK	United Kingdom
IAY-77502	05/02/2020	P003	SpinMaster Bike	799.99	GBP	C01	Cardio Equipment	SC02	Exercise Bikes	S004	Elite Gym Supply	LO	London	UK	United Kingdom
IAY-77502	05/02/2020	P014	Leg Press Machine	899.99	GBP	C02	Strength Training	SC07	Weight Machines	S004	Elite Gym Supply	LO	London	UK	United Kingdom
JOH-53793	02/01/2023	P013	Multi-Gym Station	1499.99	GBP	C02	Strength Training	SC07	Weight Machines	S004	Elite Gym Supply	LO	London	UK	United Kingdom
JOH-53793	02/01/2023	P017	Cast Iron Kettlebell 1	59.99	GBP	C03	Functional Fitness	SC09	Kettlebells	S004	Elite Gym Supply	LO	London	UK	United Kingdom
JOH-53793	02/01/2023	P003	SpinMaster Bike	799.99	GBP	C01	Cardio Equipment	SC02	Exercise Bikes	S004	Elite Gym Supply	LO	London	UK	United Kingdom
JOH-53793	02/01/2023	P002	EconoTread	699.99	GBP	C01	Cardio Equipment	SC01	Treadmills	S004	Elite Gym Supply	LO	London	UK	United Kingdom
JOH-53793	02/01/2023	P016	Heavy Duty Resistanc	39.99	GBP	C03	Functional Fitness	SC08	Resistance Bands	S004	Elite Gym Supply	LO	London	UK	United Kingdom
OQS-86591	21/05/2021	P018	Adjustable Kettlebell	149.99	USD	C03	Functional Fitness	SC09	Kettlebells	S001	FitGear Hub	NY	New York	US	United States
OQS-86591	21/05/2021	P026	Extra Thick Exercise	49.99	USD	C04	Recovery and Wellne	SC13	Yoga Mats	S001	FitGear Hub	NY	New York	US	United States
OQS-86591	21/05/2021	P005	Glide Elite Elliptical	999.99	USD	C01	Cardio Equipment	SC03	Ellipticals	S001	FitGear Hub	NY	New York	US	United States
OQS-86591	21/05/2021	P007	AquaRow 500	899.99	USD	C01	Cardio Equipment	SC04	Rowing Machines	S001	FitGear Hub	NY	New York	US	United States
EZK-98236	28/03/2021	P011	Adjustable Bench Pr	249.99	GBP	C02	Strength Training	SC06	Weight Benches	S004	Elite Gym Supply	LO	London	UK	United Kingdom
BZP-61884	13/09/2022	P006	Compact Cross Train	549.99	USD	C01	Cardio Equipment	SC03	Ellipticals	S003	Strength Central	CH	Chicago	US	United States
BZP-61884	13/09/2022	P023	PulseRelief Massage	199.99	USD	C04	Recovery and Wellne	SC12	Massage Guns	S003	Strength Central	CH	Chicago	US	United States
BZP-61884	13/09/2022	P003	SpinMaster Bike	799.99	USD	C01	Cardio Equipment	SC02	Exercise Bikes	S003	Strength Central	CH	Chicago	US	United States

2. Creating the Dim tables

a.

store_id	store_name	city_id	country_id
S001	FitGear Hub	NY	US
S002	Muscle Mart	LA	US
S003	Strength Central	CH	US
S004	Elite Gym Supply	LO	UK
S005	Active Life Outfitters	MA	UK

b.

city_id	city_name
MA	Manchester
LA	Los Angeles
NY	New York
CH	Chicago
LO	London

c.

country_id	country
UK	United Kingdom
US	United States

d.

category_id	category_name
C01	Cardio Equipment
C02	Strength Training
C03	Functional Fitness
C04	Recovery and Wellness

e.

subcategory_id	subcategory_name
SC01	Treadmills
SC02	Exercise Bikes
SC03	Ellipticals
SC04	Rowing Machines
SC05	Free Weights
SC06	Weight Benches
SC07	Weight Machines
SC08	Resistance Bands
SC09	Kettlebells
SC10	Suspension Trainers
SC11	Foam Rollers
SC12	Massage Guns
SC13	Yoga Mats

f.

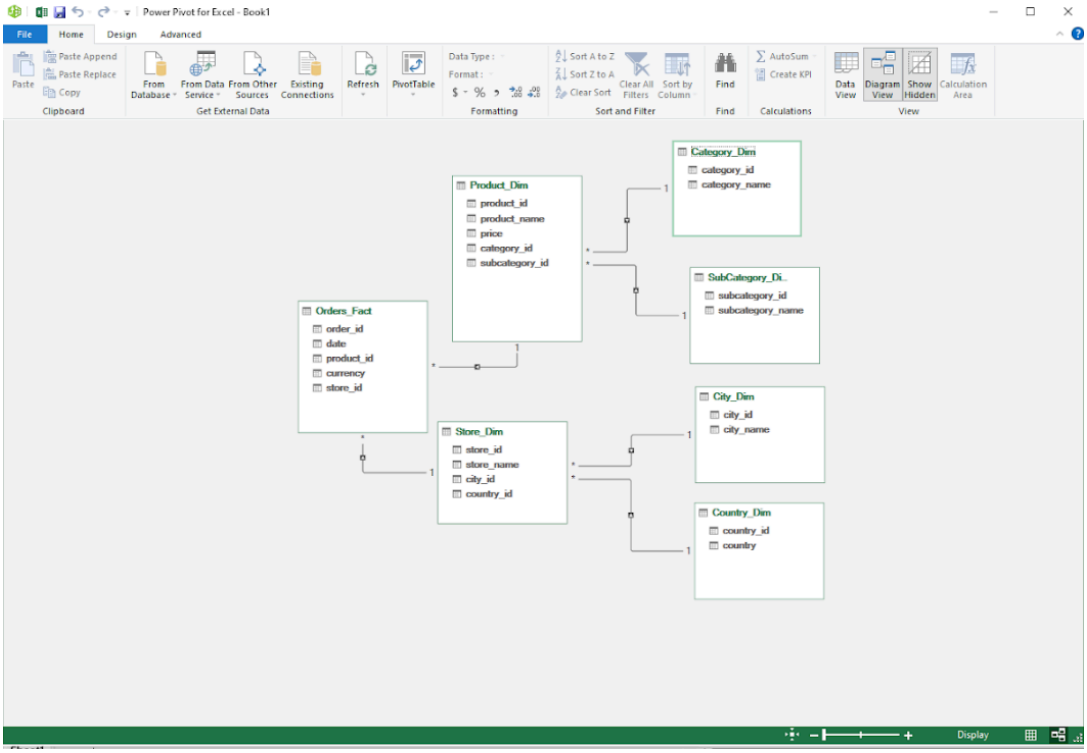
product_id	product_name	price	category_id	subcategory_id
P002	EconoTread	699.99	C01	SC01
P020	Door Anchor Straps	24.99	C03	SC10
P012	Olympic Flat Bench	179.99	C02	SC06
P005	Glide Elite Elliptical	999.99	C01	SC03
P004	Recumbent Cycler	599.99	C01	SC02
P003	SpinMaster Bike	799.99	C01	SC02
P016	Heavy Duty Resistance Tubes	39.99	C03	SC08
P024	Mini Percussion Massager	79.99	C04	SC12
P019	Pro Suspension Trainer	129.99	C03	SC10
P017	Cast Iron Kettlebell 16kg	59.99	C03	SC09
P026	Extra Thick Exercise Mat	49.99	C04	SC13
P021	Recovery Roller Deluxe	39.99	C04	SC11
P023	PulseRelief Massage Gun	199.99	C04	SC12
P011	Adjustable Bench Pro	249.99	C02	SC06
P022	Trigger Point Grid Roller	59.99	C04	SC11
P015	FlexForce Band Kit	49.99	C03	SC08
P001	Pro Runner 3000	1299.99	C01	SC01
P018	Adjustable Kettlebell	149.99	C03	SC09
P025	Premium Yoga Mat	69.99	C04	SC13
P014	Leg Press Machine	899.99	C02	SC07
P013	Multi-Gym Station	1499.99	C02	SC07
P007	AquaRow 500	899.99	C01	SC04
P006	Compact Cross Trainer	549.99	C01	SC03
P008	AirRow Pro	1099.99	C01	SC04
P010	Cast Iron Plate Set	199.99	C02	SC05
P009	PowerLift Dumbbell Set	299.99	C02	SC05

3. Creating the Fact Table

a.

order_id	date	product_id	currency	store_id
FVJ-53564	2/27/2023	P010	GBP	S004
XRR-35528	9/8/2022	P010	GBP	S004
TVI-73860	7/12/2022	P010	GBP	S004
NFL-15231	9/11/2020	P010	GBP	S004
DCH-18218	12/26/2023	P010	GBP	S004
DCH-18218	12/26/2023	P010	GBP	S004
ZVZ-57333	1/1/2022	P010	GBP	S004
HKO-36433	10/11/2023	P010	GBP	S004
WBE-25703	11/27/2023	P010	GBP	S004
QPO-09724	8/15/2021	P010	GBP	S004
OLK-42485	11/5/2020	P010	GBP	S004
BMG-47274	5/9/2023	P010	GBP	S004
DXH-58332	7/14/2021	P010	GBP	S004
EYF-66445	3/24/2022	P010	GBP	S004
TNM-60609	10/6/2023	P010	GBP	S004
EJP-80058	6/12/2020	P010	GBP	S004
KWQ-50566	5/8/2022	P010	GBP	S004
FEM-34624	10/15/2021	P010	GBP	S004
MMO-64501	5/14/2021	P010	GBP	S004
XGR-47222	7/19/2021	P010	GBP	S004
YDP-93085	11/26/2020	P010	GBP	S004
ZLM-96556	3/14/2020	P010	GBP	S004
GSN-33629	5/21/2021	P010	GBP	S004
EOL-39350	3/31/2021	P010	GBP	S004
SXE-52173	5/11/2022	P010	GBP	S004
SFM-82596	12/7/2022	P010	GBP	S004
LWD-59534	5/6/2020	P010	GBP	S004
UVU-42922	7/6/2022	P010	GBP	S004
VMX-86584	3/17/2023	P010	GBP	S004
DSG-40374	6/10/2021	P010	GBP	S004
FJR-83386	10/12/2022	P010	GBP	S004
EUO-62039	8/28/2023	P010	GBP	S004
DVA-99133	10/28/2021	P010	GBP	S004
FED-26818	3/12/2020	P010	GBP	S004
WYB-59430	6/18/2021	P010	GBP	S004
ZWT-48103	9/2/2021	P010	GBP	S004
GWP-57893	4/24/2020	P010	GBP	S004
HWJ-83259	5/3/2020	P010	GBP	S004
ZMQ-21091	2/10/2023	P010	GBP	S004
ZMQ-21091	2/10/2023	P010	GBP	S004
LOE-79911	10/22/2021	P010	GBP	S004
FYI-17403	8/29/2021	P010	GBP	S004
DTS-84961	12/11/2021	P010	GBP	S004
DTS-84961	12/11/2021	P010	GBP	S004

Store_Dim | City_Dim | Country_Dim | Category_Dim | SubCategory_Dim | Product_Dim | Orders_Fact

4. The model Diagram

a.